SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204
Estd:1980

COMPUTER SCIENCE & ENGINEERING
(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from **2019-2020**Admitted Batch onwards**)**

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 CS 3101	Data Warehousing and Data Mining	PC	3	3	--	--	25	75	100
B19 CS 3102	Computer Networks	PC	3	3	--	--	25	75	100
B19 CS 3103	Formal Languages and Automata Theory	PC	3	3	--	--	25	75	100
B19 CS 3104	Web Technologies	PC	3	3	--	--	25	75	100
# PE-I	Professional Elective-I	PE	3	3	--	--	25	75	100
B19 CS 3110	Computer Networks Lab	PC	1.5	--	--	3	20	30	50
B19 CS 3111	Web Technologies Lab	PC	1.5	--	--	3	20	30	50
B19 CS 3112	Data Mining Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability Skills-I	MC	0	3	--	--	--	--	--
B19 MC 3103	Advanced Coding	MC	0	--	--	3	--	--	--
		TOTAL	**19.5**	**18**	**--**	**12**	**185**	**465**	**650**

	Course Code	Course
#PE-I	B19 CS 3105	Computer Graphics
	B19 CS 3106	Principles of Programming Languages
	B19 CS 3107	Scripting Languages
	B19 CS 3108	Advanced Computer Architecture
	B19 CS 3109	Software Testing Methodologies

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3101	**PC**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

DATA WAREHOUSING AND DATA MINING

(For CSE)

Course Objectives:

1.	To understand data warehouse concepts, architecture and the evolution of data warehousing and data mining techniques.
2.	To understand the types of data, their characteristics, cleaning, and transformation of data
3.	To learn the principles of statistics, information theory, databases, machine learning and other areas for design and implementation of data mining techniques.
4.	To understand and apply various association rule mining, classification and clustering techniques. Course

Course Outcomes:
At the end of the course Students will be able to

1.	Identify the importance and applications of Data Mining and able to interpret the data.	K3
2.	Explain the concepts of data warehousing and OLAP technology and apply data pre-processing techniques on raw data to make it suitable for data mining.	K3
3.	Formulate and apply association rule mining algorithms and their performance evaluation metrics on sample datasets	K3
4.	Formulate and apply classification, prediction and their respective performance evaluation metrics on sample datasets	K3
5.	Illustrate, apply and compare partitional, hierarchical, density based and grid based clustering algorithms	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Data Mining :** Evolution of I T into DBMS, Motivation and importance of Data Warehousing and Data Mining, Kinds of Patterns, Technologies, Basic Data Analytics: Data Objects and Attributes Types, Statistical Descriptions of Data, Data Visualization, Estimating Data Similarity and Dissimilarity, Major Issues in Data Mining, Data Mining Applications..
UNIT-II (10 Hrs)	**Data Preprocessing and OLAP Technology:** Data Cleaning, Data Integration, Data Reduction, Data Transformation, Discretization and Concept Hierarchy Generation. Basic Concepts of Data warehouse, Data Modelling using Cubes and OLAP, DWH Design and usage, Implementation using Data Cubes and OLAPs.
UNIT-III (10 Hrs)	**Mining Frequent Patterns Based on Associations and Correlations:** Basic Concepts, Frequent Item set Mining Methods: A priori Algorithm, Association Rule Generation, Improvements to A Priori, FP- Growth Approach, Mining Frequent Patterns using Data Warehousing and Data Mining Data Formats, Mining Closed and Max Patterns, Pattern Evaluation Methods, mining in multilevel, multidimensional space.

UNIT-IV (10 Hrs)	**Classification & Prediction:** Basic Concepts, Decision Tree Induction, Bayes Classification, Rule- Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy Advanced Methods: Classification by Back Propagation.
UNIT-V (10 Hrs)	**Cluster Analysis:** Basic Concepts and issues in clustering, Types of Data in Cluster Analysis, Partitioning Methods-K-Means, K-Medoids, Hierarchical Methods-Agglomerative versus DivisiveDistance Measures in Algorithmic Methods, Birch, Density Based Methods- DBSCAN, OPTICS, Grid Based Methods-STING.

Text Books:	
1.	Data Mining- Concepts and Techniques by Jiawei Han, MichelineKamber and Jian Pei –Morgan Kaufmann publishers –--3 rd edition, June 2012
Reference Books:	
1.	Introduction to Data Mining, Adriaan, Addison Wesley Publication, 2005
2.	Data Mining Techniques, A.K.Pujari, University Press, 2013
3.	Introduction to Data mining by Vipin Kumar, Pang-Ning Tan, Michael Steinbach - Pearson Edu publishers, 2016
4.	Data Mining –Introductory and Advanced concepts by Margarett Dunham -- Pearson Education publishers, 2006
e-Resources:	
1.	NPTEL course by Prof. PabitraMitra, https://onlinecourses.nptel.ac.in/noc21_cs06/preview
2.	Cluster Analysis in Data Mining, https://www.coursera.org/learn/cluster-analysis? specialization=data-mining

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3102	PC	3	0	0	3	25	75	3 Hrs.

COMPUTER NETWORKS

(For CSE)

Course Objectives:

1.	Study the basic taxonomy and terminology of the computer networking and enumerate the layers of OSI model and TCP/IP model
2.	Study data link layer concepts, design issues, and protocols
3.	Gain core knowledge of Network layer routing protocols and IP addressing
4.	Study Session layer design issues, Transport layer services, and protocols
5.	Acquire knowledge of Application layer and Presentation layer paradigms and protocols

Course Outcomes:
At the end of the course Students will be able to

1.	Illustrate the OSI and TCP/IP reference model	K2
2.	Able to identify the MAC layer protocols and LAN technology	K3
3.	Demonstrate various applications using internet protocol	K3
4.	Summarize various routing and congestion control algorithms	K2
5.	Make use of various application layer protocols	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** History and development of computer networks, Basic Network Architectures: OSI reference model, TCP/IP reference model, and Networks topologies, types of networks (LAN, MAN, WAN, circuit switched, packet switched, message switched, extranet, intranet, Internet, wired, wireless).
UNIT-II (10 Hrs)	**Physical layer:** Line encoding, block encoding, scrambling, modulation demodulation (both analog and digital), errors in transmission, multiplexing (FDM, TDM), Different types of transmission media. Data Link Layer services: framing, error control, flow control, medium access control. Error & Flow control mechanisms: stop and wait, Go back N and selective repeat. MAC protocols: Aloha, slotted aloha, CSMA, CSMA/CD, and CSMA/CA.
UNIT-III (10 Hrs)	**Local Area Network Technology:** Token Ring. Error detection (Parity, CRC), Ethernet, Fast Ethernet, Gigabit Ethernet, Personal Area Network: Bluetooth and Wireless Communications Standard: Wi-Fi (802.11).
UNIT-IV (10 Hrs)	**Network layer:** Internet Protocol, IPv6, ARP, DHCP, ICMP, Routing algorithms: Distance vector, Link state, Metrics, Inter-domain routing. Subnetting, Supernetting, Classless addressing, Network Address Translation.
UNIT-V (10 Hrs)	**Transport layer:** UDP, TCP. Connection establishment and termination, sliding window, flow and congestion control, timers, retransmission. Application Layer. Network Application services and protocols including e-mail, www, DNS, SMTP, IMAP, FTP, TFTP, Telnet, BOOTP, HTTP, IPSec, Firewalls.

Text Books:	
1	Computer Networks, Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 5th edition, 2013.
2	Data Communication and Networking , Behrouz A. Forouzan, McGraw Hill, 5th Edition,2012
Reference Books:	
1.	Computer Networks: A Systems Approach, LL Peterson, BS Davie, Morgan-Kauffman, 5th Edition, 2011.
2.	Computer Networking: A Top-Down Approach JF Kurose, KW Ross, Addison-Wesley, 5th Edition, 2009.
3.	Data and Computer Communications , William Stallings , Pearson , 8th Edition, 2007
4.	Computer Networks, Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 7th edition.
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105183
2.	https://nptel.ac.in/courses/106/106/106106091/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3103	**PC**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

FORMAL LANGUAGES AND AUTOMATA THEORY

(For CSE)

Course Objectives:

1.	To learn how to design Automata's as Acceptors, Verifiers and Translators
2.	To learn fundamentals of Regular and Context Free Grammars and Languages
3.	To understand the relation between different Languages and Automata
4.	To learn how to design PDA and TM

Course Outcomes:
At the end of the course Students will be able to

1.	Construct DFA, NFA, NFA with Є-Transitions, Mealy and Moore machines	K3
2.	Illustrate regular expressions, equivalence of FA and Regular expressions and properties of Regular languages.	K2
3.	Construct CFG, Regular grammars, simplification and Normal forms in CFG	K3
4.	Illustrate properties of CFL and Construct PDA	K3
5.	Summarize decidable and un-decidable problems and Construct of TM	K3

SYLLABUS

UNIT-I (10 Hrs)	Why Study Automata Theory? Central Concepts of Automata Theory, Introduction to DFA and NFA, Acceptance of a String by a DFA, Acceptance of a String by a NFA, Design of DFAs, Design of NFAs, Conversion of NFA to DFA, Introduction to NFA with Є-Transitions, Conversion of NFA with Є-Transitions to NFA without Є-Transitions. Minimization of DFA, Introduction to Mealy and Moore Machines, Design of Mealy and Moore Machines, Conversion of Mealy to Moore Machines and Moore to Mealy Machines, Applications and Limitations of Finite Automata.
UNIT-II (10 Hrs)	Introduction to Regular Expressions, Regular Sets, Identity Rules, Equivalence of two Regular Expressions, Conversion of Regular Expression to NFA with Є-Transitions, Conversion of DFA to Regular Expression. Pumping Lemma of Regular Languages, Applications of pumping lemma, Closure Properties of Regular Languages, Applications of Regular Expressions
UNIT-III (10 Hrs)	Chomsky Hierarchy, Regular Grammar, Left-Linear Grammar, Right-Linear Grammar, Conversion of Finite Automata to Regular Grammars and Regular Grammars to Finite Automata, Context Free Grammar, Construction of CFGs for Languages, Determining language of the grammar, Leftmost and Rightmost Derivations, Parse Trees. Ambiguous Grammars, Simplification of Context Free Grammars (Elimination of Useless Symbols, Є-Productions and Unit Productions), Normal Forms (Chomsky Normal Form and Greibach Normal Form).

UNIT-IV (10 Hrs)	Pumping Lemma for CFL, Applications of pumping lemma for CFL, Closure Properties of CFL, Applications of Context Free Grammars, Introduction to Pushdown Automata, Model, Graphical Notation, Instantaneous Description, Language Acceptance of Pushdown Automata (Acceptance by empty stack and final state), Design of Pushdown Automata for CFL. Deterministic and Non–Deterministic Pushdown Automata, Conversion of Pushdown Automata to Context Free Grammars, Conversion of Context Free Grammars to Pushdown Automata, Application of Pushdown Automata.
UNIT-V (10 Hrs)	Introduction to Turning Machine, Representation of Turning Machines (Instantaneous Descriptions, Transition Tables and Transition Diagrams), Design of Turning Machines. Types of Turning Machines, Church's Thesis, Universal Turning Machine, Introduction to Decidable and Un-decidable Problems, Halting Problem of Turning Machines, Post's Correspondence Problem, Modified Post's Correspondence Problem, Introduction to Classes of P and NP, NP-Hard and NP-Complete Problems

Text Books:

1	Introduction to Automata Theory, Languages and Computation, J. E. Hopcroft, R. Motwani and J. D. Ullman, 3rd Edition, Pearson, 2008
2	Theory of Computer Science-Automata, Languages and Computation, K. L. P. Mishra and N. Chandrasekharan, 3rd Edition, PHI, 2007

Reference Books:

1.	Elements of Theory of Computation, Lewis H.P. &Papadimition C.H., Pearson /PHI
2.	Theory of Computation, V. Kulkarni, Oxford University Press, 2013
3.	Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill, 2014

e-Resources:

1.	https://nptel.ac.in/courses/106/104/106104028

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3104	**PC**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

WEB TECHNOLOGIES

(For CSE)

Course Objectives:

1.	Able to develop static web pages using HTML, CSS and XML.
2.	Able to develop interactive web pages using Java Script.
3.	Able to write a well formed or valid XML document and XML Schema.
4.	Able to build interactive web applications using AJAX.
5.	Able to understand basics of Server side scripting using PHP.

Course Outcomes:
At the end of the course Students will be able to

1.	Illustrate the basic concepts of HTML and CSS & apply those concepts to design static web pages	K3
2.	Identify and understand various concepts related to dynamic web pages and validate them using JavaScript	K2
3.	Outline the concepts of Extensible mark-up language & AJAX	K2
4.	Illustrate the basic concepts of PHP and implement those concepts	K3
5.	Construct and deploy secure, usable database driven web applications using PHP	K3

SYLLABUS

UNIT-I (10 Hrs)	**HTML:** Basic Syntax, Core Elements, Links and Addressing, Images, Iframe Images, Text, Hyper Text Links, Colors and Background, Lists, Tables and Layouts, Frames, Forms GET and POST methods.
UNIT-II (10 Hrs)	**Dynamic HTML. CSS:** Cascading style sheets, Levels of Style Sheets, Style Specification Formats, Selector Forms. **JavaScript** - Introduction to JavaScript, Objects, Primitives Operations and Expressions, Control Statements, Arrays, Functions, Pattern Matching using Regular Expressions.
UNIT-III (10 Hrs)	**Working with XML**: Document type Definition (DTD), XML schemas, Document object model, Parsers- DOM and SAX. **AJAX A New Approach:** Introduction to AJAX, Basics of AJAX, XML Http Request Object, Integrating PHP and AJAX.
UNIT-IV (10 Hrs)	**PHP Programming:** Introduction to PHP, Creating PHP script, Running PHP script. Working with variables and constants: Using variables, Using constants, Data types, Operators. Controlling program flow: Conditional statements, Control statements, Arrays, functions.

UNIT-V (10 Hrs)	MYSQL Installation, Accessing MYSQL using PHP, Form Handling, Cookies, Session Tracking, Tables, inserting data into Tables, Selecting Data from a Table, Updating Table, Deleting data from Table, Webpage creation.

Text Books:

1	Programming the World Wide Web, 7th Edition Robet W Sebesta, Pearson, 2013.
2	WebTechnologies,1st Edition 7th impression, Uttam K Roy, Oxford, 2012.
3	Pro Mean Stack Development,1st Edition, ELadElrom, A press O'Reilly,2016
4	Java Script & jQuery the missing manual, 2nd Edition, David sawyer mcfarland, O'Reilly, 2011.
5	Web Hosting for Dummies, 1st Edition, Peter Pollock, John Wiley & Sons, 2013.
6	RESTful web services, 1st Edition, Leonard Richardson, Ruby, O'Reilly, 2007.

Reference Books:

1.	Ruby on Rails Up and Running, Lightning fast Web development, 1st Edition, Bruce Tate, Curt Hibbs, Oreilly, 2006.
2.	Programming Perl, 4th Edition, Tom Christiansen, Jonathan Orwant, O'Reilly, 2012.
3.	Web Technologies, HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, 1st Edition, Dream Tech, 2009.
4.	An Introduction to Web Design, Programming, 1st Edition, Paul S Wang, Sanda S Katila, Cengage Learning, 2003.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3105	PE	3	0	0	3	25	75	3 Hrs.

COMPUTER GRAPHICS

(Professional Elective-I)

(For CSE)

Course Objectives:

1.	Provide a comprehensive introduction to computer graphics leading to the ability to understand contemporary terminology, technology, progress and trends.
2.	Design of algorithms for digitization of graphic primitives.
3.	Apply computer graphics techniques for two-dimensional and three-dimensional transformations.
4.	Visualize viewing transformations.
5.	Demonstrate working of I/O devices.

Course Outcomes:

At the end of the course Students will be able to

1.	Summarize the application areas of computer graphics and the working of I/O devices	K2
2.	Implement algorithms for scan converting graphic primitives in a graphic package.	K3
3.	Apply direct and indirect methods for two-dimensional transformations using matrices.	K3
4.	Construct three-dimensional geometric transformations using matrices	K3
5.	Produce two-dimensional and three dimensional viewing transformations	K3

SYLLABUS

UNIT-I (10 Hrs)	Overview **of Graphics Systems:** Applications of Computer Graphics-Graphical User Interfaces-Video Display Devices-Raster Scan Systems-Random Scan Systems-Graphics Monitors and Workstations -Input Devices- Logical Classification of Input Devices-Hard Copy Devices- Graphics Software-Overview of C- Graphics, Open GL and PHIGS.
UNIT-II (10 Hrs)	**Output Primitives and it Attributes:** Points and Lines-Line Drawing Algorithms-Circle Generating Algorithms- Parallel Line Algorithms-Functions in C-Graphics for Output Primitives-Color and Gray Scale Levels - Boundary Fill Algorithm- Flood Fill Algorithm -Anti-aliasing Techniques.
UNIT-III (10 Hrs)	**Two-Dimensional Geometric Transformations:**Basic Transformations- Matrix Representations- Homogeneous Coordinates- Composite Transformations- Reflection-Shear- Transformations between Coordinate Systems- Affine Transformations- Raster Methods for Transformations.
UNIT-IV (10 Hrs)	**Two-Dimensional Viewing:** The Viewing Pipeline-Viewing Coordinate Reference Frame-Window-to-Viewport Coordinate Transformation-Clipping Operations-Point Clipping-Line Clipping-Polygon Clipping-Curve Clipping- Text and Exterior Clipping
UNIT-V (10 Hrs)	**Three-Dimensional Geometric Transformations and Viewing:** **3D Transformations:** Translation- Rotation- Scaling- Reflection -Shear- Composite

	Transformations-Modeling and Coordinate Transformations. **3D Display Methods:** Spline Representations-Natural Cubic Spline- Bézier Curves and Surfaces **3D Viewing:** 3D Viewing Pipeline- Viewing Coordinates- View Volumes- General Projections

Text Books:	
1.	Computer Graphics C Version, Donald Hearn& M. Pauline Baker, 2nd Edition, Pearson Education, 2014.
2.	Computer Graphics with Open-GL, Donald Hearn, M. Pauline Baker& Warren Carithers, 4th Edition, Pearson Education, 2014.
Reference Books:	
1.	Procedural Elements for Computer Graphics,David F. Rogers, Indian Edition, Tata McGraw Hill Education, 2017
2.	Computer Graphics, Zhigang Xiang and Roy A. Plastock, 2nd Edition (Indian Edition) McGraw-Hill Education, 2015
3.	Computer Graphics: Principles and Practice, John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, Kurt Akeley, 3rd Edition, Addison-Wesley Professional, 2013.
4.	Mathematical and computer programming techniques for computer graphics, PeterComninos, Springer, 2010.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3106	**PE**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

PRINCIPLES OF PROGRAMMING LANGUAGES

(Professional Elective-I)

(For CSE)

Course Objectives:

1.	To understand and describe syntax and semantics of programming languages
2.	To understand data, data types, and basic statements
3.	To understand call-return architecture and ways of implementing them
4.	To understand object-orientation, concurrency, and event handling in programming languages
5.	To develop programs in non-procedural programming paradigms

Course Outcomes:
At the end of the course Students will be able to

1.	Describe the syntax and semantics of programming languages to gain practical knowledge in lexical analysis and parsing phases of a compiler	K3
2.	Make use of different constructs in programming languages with merits and demerits.	K3
3.	Design and implement sub programs in various programming languages	K3
4.	Develop the knowledge on different programming language features like object-orientation, concurrency, exception handling and event handling	K4
5.	Analyze functional paradigm and ability to write small programs using Scheme and ML and Develop programs logic paradigm and ability to write small programs using Prolog	K4

SYLLABUS

UNIT-I (10 Hrs)	**Syntax and semantics:** Evolution of programming languages, describing syntax, context, free grammars, attribute grammars, describing semantics, lexical analysis, parsing, recursive - decent bottom - up parsing.
UNIT-II (10 Hrs)	**Data, data types, and basic statements:** Names, variables, binding, type checking, scope, scope rules, lifetime and garbage collection, primitive data types, strings, array types, associative arrays, record types, union types, pointers and references, Arithmetic expressions, overloaded operators, type conversions, relational and Boolean expressions, assignment statements, mixed mode assignments, control structures – selection, iterations, branching, guarded Statements.
UNIT-III (10 Hrs)	**Subprograms and implementations:** Subprograms, design issues, local referencing, parameter passing, overloaded methods, generic methods, design issues for functions, semantics of call and return, implementing simple subprograms, stack and dynamic local variables, nested subprograms, blocks, dynamic scoping.
UNIT-IV (10 Hrs)	**Object- orientation, concurrency, and event handling**: Object – orientation, design issues for OOP languages, implementation of object, oriented constructs, concurrency, semaphores, Monitors, message passing, threads, statement level concurrency, exception handling, event handling.

UNIT-V (10 Hrs)	**Functional programming languages**: Introduction to lambda calculus, fundamentals of functional programming languages, Programming with Scheme, Programming with ML Logic programming languages: Introduction to logic and logic programming, Programming with Prolog, multi - paradigm languages.

Text Books:	
1.	Robert W. Sebesta, "Concepts of Programming Languages", Tenth Edition, Addison Wesley, 2012.
2.	Programming Languages, Principles & Paradigms, 2ed, Allen B Tucker, Robert E Noonan, TMH.
Reference Books:	
1.	R. Kent Dybvig, "The Scheme programming language", Fourth Edition, MIT Press, 2009.
2.	Jeffrey D. Ullman, "Elements of ML programming", Second Edition, Prentice Hall, 1998.
3.	Richard A. O'Keefe, "The craft of Prolog", MIT Press, 2009.
4.	W. F. Clocksin and C. S. Mellish, "Programming in Prolog: Using the ISO Standard", Fifth Edition, Springer, 2003.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3107	PE	3	0	0	3	25	75	3 Hrs.

SCRIPTING LANGUAGES

(Professional Elective-I)

(For CSE)

Course Objectives:

1.	Understand the concepts of scripting languages for developing web based projects
2.	Illustrates object oriented concepts like PHP, PYTHON, PERL
3.	Create database connections using PHP and build the website for the world
4.	Demonstrate IP address for connecting the web servers
5.	Analyze the internet ware application, security issues and frame works for application

Course Outcomes:
At the end of the course Students will be able to

1.	Understand the differences between scripting languages	K2
2.	Create PHP authentication Methodology for security issue..	K4
3.	Identify PHP encryption functions and Mcrypt Package	K2
4.	Explain syntax and variables in TCL	K2
5.	Gain some fluency programming in Ruby, JavaScript, Perl, Python, and related languages	K3
6.	Understand object oriented concepts in python	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to PERL and Scripting:** Scripts and Programs, Origin of Scripting, Scripting Today,Characteristics of Scripting Languages, Uses for Scripting Languages, Web Scripting, and the universe of Scripting Languages. PERL- Names and Values, Variables, Scalar Expressions, Control Structures, arrays, list, hashes, strings, pattern and regular expressions, subroutines.
UNIT-II (10 Hrs)	**Advanced PERL:** Finer points of looping, pack and unpack, file system, eval, data structures,packages, modules, objects, interfacing to the operating system, Creating Internet ware applications, Dirty Hands Internet Programming, security Issues.PHP Basics: PHP Basics- Features, Embedding PHP Code in your Web pages, Outputting the data tothe browser, Data types, Variables, Constants, expressions, string interpolation, control structures, Function, Creating a Function, Function Libraries, Arrays, strings and Regular Expressions.
UNIT-III (10 Hrs)	**Advanced PHP Programming:** PHP and Web Forms, Files, PHP Authentication and Methodologies Hard Coded, File Based, Database Based, IP Based, Login Administration, Uploading Files with PHP, Sending Email using PHP, PHP Encryption Functions, the M crypt package, Building Web sites for the World.

UNIT-IV (10 Hrs)	**TCL:** TCL Structure, syntax, Variables and Data in TCL, Control Flow, Data Structures, input/output, procedures , strings , patterns, files, Advance TCL- eval, source, exec and uplevel commands, Name spaces, trapping errors, event driven programs, making applications internet aware, Nuts and Bolts Internet Programming, Security Issues, C Interface. Tk-Visual Tool Kits, Fundamental Concepts of Tk, Tk by example, Events and Binding , Perl-Tk.
UNIT-V (10 Hrs)	**Python:** Introduction to Python language, python-syntax, statements, functions, Built-in-functionsand Methods, Modules in python, Exception Handling. Integrated Web Applications in Python – Building Small, Efficient Python Web Systems, Web Application Framework.

Text Books:	
1.	The World of Scripting Languages, David Barron, Wiley Publications.
2.	Python Web Programming, Steve Holden and David Beazley, New Riders Publications.
3.	Beginning PHP and MySQL, 3rd Edition, Jason Gilmore, Apress Publications (Dream tech)
Reference Books:	
1.	Open Source Web Development with LAMP using Linux, Apache, MySQL, Perl and PHP, J.Lee andB.Ware (Addison Wesley) Pearson Education. Programming Python,M.Lutz,SPD.
2.	PHP 6 Fast and Easy Web Development, Julie Meloni and Matt Telles, Cengage Learning Publications.
3.	Tcl and the Tk Tool kit, Ousterhout, Pearson Education.
4.	PHP and MySQL by Example, E.Quigley, Prentice Hall (Pearson).
5.	Perl Power, J.P.Flynt, Cengage Learning.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3108	**PE**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

ADVANCED COMPUTER ARCHITECTURE

(Professional Elective-I)

(For CSE)

Course Objectives:

1.	Understand the Concept of Parallel Processing and its applications
2.	Implement the Hardware for Arithmetic Operations
3.	Analyze the performance of different scalar Computers
4.	Develop the Pipelining Concept for a given set of Instructions
5.	Distinguish the performance of pipelining and non pipelining environment in a processor

Course Outcomes:
At the end of the course Students will be able to

1.	Illustrate the types of computers, and new trends and developments in computer architecture	K2
2.	Outline pipelining, instruction set architectures, memory addressing	K3
3.	Apply ILP using dynamic scheduling, multiple issue, and speculation	K3
4.	Illustrate the various techniques to enhance a processors ability to exploit Instruction-level parallelism (ILP), and its challenges	K2
5.	Apply multithreading by using ILP and supporting thread-level parallelism (TLP)	K3

SYLLABUS

UNIT-I (10 Hrs)	**Computer Abstractions and Technology:** Introduction, Eight Great Ideas in Computer Architecture, Below Your Program, Under the Covers, Technologies for Building Processors and Memory, Performance, The Power Wall, The Sea Change: The Switch from Uni-processors to Multiprocessors, Benchmarking the Intel Core i7, Fallacies and Pitfalls.
UNIT-II (10 Hrs)	**Instructions:** Language of the Computer: Operations of the Computer Hardware, Operands of the Computer Hardware, Signed and Unsigned Numbers, Representing Instructions in the Computer, Logical Operations, Instructions for Making Decisions, Supporting Procedures in Computer Hardware, Communicating with People, MIPS Addressing for 32-Bit Immediates and Addresses, Parallelism and Instructions: Synchronization, Translating and Starting a Program, A C Sort Example to Put It All Together, Arrays versus Pointers, ARMv7 (32-bit) Instructions, x86 Instructions, ARMv8 (64-bit) Instructions.
UNIT-III (10 Hrs)	**Arithmetic for Computers:** Introduction, Addition and Subtraction, Multiplication, Division, Floating Point, Parallelism and Computer Arithmetic: Subword Parallelism, Streaming SIMD Extensions and Advanced Vector Extensions in x86, Subword Parallelism and Matrix Multiply.
UNIT-IV (10 Hrs)	**The Processor:** Introduction, Logic Design Conventions, Building a Datapath, A Simple Implementation Scheme, An Overview of Pipelining, Pipelined Datapath and Control, Data Hazards: Forwarding versus Stalling, Control Hazards, Exceptions, Parallelism via Instructions, The ARM Cortex-A8 and Intel Core i7 Pipelines.

UNIT-V (10 Hrs)	**Large and Fast:** Exploiting Memory Hierarchy: Introduction, Memory Technologies, The Basics of Caches, Measuring and Improving Cache Performance, Dependable Memory Hierarchy, Virtual Machines, Virtual Memory, A Common Framework for Memory Hierarchy, Using a Finite-State Machine to Control a Simple Cache, Parallelism and Memory Hierarchies: Cache Coherence, Parallelism and Memory Hierarchy: Redundant Arrays of Inexpensive Disks, Advanced Material: Implementing Cache Controllers, The ARM Cortex-A8 and Intel Core i7 Memory Hierarchies

Text Books:	
1.	Computer Organization and Design: The hardware and Software Interface, David A Patterson, John L Hennessy, 5th edition, MK.
2.	Computer Architecture and Parallel Processing – Kai Hwang, Faye A.Brigs, Mc Graw Hill.
Reference Books:	
1.	Modern Processor Design: Fundamentals of Super Scalar Processors, John P. Shen and Miikko H. Lipasti, Mc Graw Hill.
2.	Advanced Computer Architecture – A Design Space Approach – DezsoSima, Terence Fountain, Peter Kacsuk , Pearson.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3109	**PE**	**3**	**0**	**0**	**3**	**25**	**75**	**3 Hrs.**

SOFTWARE TESTING METHODOLOGIES

(Professional Elective-I)

(For CSE)

Course Objectives:

1.	To study fundamental concepts in software testing and discuss various software testing issues and solutions in software unit, integration, regression and system testing
2.	To learn how to plan a test project, design test cases and data, conduct testing, manage software problems and defects, generate a test report
3.	To expose the advanced software testing concepts such as object-oriented software testing methods, web-based and component-based software testing
4.	To understand software test automation problems and solutions
5.	To learn how to write software test documents and communicate with engineers in various forms

Course Outcomes:

At the end of the course Students will be able to

1.	Identify and understand various software testing problems, apply software testing knowledge and engineering methods and solve these problems by designing and selecting software test models, criteria, strategies, and methods	K2
2.	Design and conduct a software test process for a software project	K4
3.	Analyze the needs of software test automation	K4
4.	Use various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects	K2
5.	Understand and gain knowledge on contemporary issues in software testing, such as component-based, web based and object oriented software testing problems	K2
6.	Write test cases for given software to test it before delivery to the customer and write test scripts for both desktop and web based applications	K3

SYLLABUS

UNIT-I (10 Hrs)	**Software Testing:** Introduction, Evolution, Myths & Facts, Goals, Psychology, definition, Model for testing, Effective Vs Exhaustive Software Testing. Software Testing Terminology and Methodology: Software Testing Terminology, Software Testing Life Cycle, Software Testing Methodology. Verification and Validation: Verification & Validation Activities, Verification, Verification of Requirements, High level and low level designs, verifying code, Validation
UNIT-II (10 Hrs)	**Dynamic Testing-Black Box testing techniques:** Boundary Value Analysis, Equivalence class Testing, State Table based testing, Decision table based testing, Cause-Effect Graphing based testing, Error guessing White-Box Testing: need, Logic Coverage criteria, Basis Path testing, Graph matrices, Loop testing, data flow testing, mutation testing

UNIT-III (10 Hrs)	**Static Testing: Inspections, Structured Walkthroughs, Technical Reviews Validation activities:** Unit testing, Integration Testing, Function testing, system testing, acceptance testing Regression testing: Progressives Vs regressive testing, Regression test ability, Objectives of regression testing, Regression testing types, Regression testing techniques.
UNIT-IV (10 Hrs)	**Efficient Test Suite Management:** Growing nature of test suite, Minimizing the test suite and its benefits, test suite prioritization, Types of test case prioritization, prioritization techniques, measuring the effectiveness of a prioritized test suite Software Quality Management: Software Quality metrics, SQA models Debugging: process, techniques, correcting bugs.
UNIT-V (10 Hrs)	**Automation and Testing Tools:** Need for automation, categorization of testing tools, selection of testing tools, Cost incurred, Guidelines for automated testing, overview of some commercial testing tools such as Win Runner, Load Runner, Jmeter and JUnit . Test Automation using Selenium tool.Testing Object Oriented Software: basics, Object oriented testing ,Testing Web based Systems: Challenges in testing for web based software, quality aspects, web engineering, testing of web based systems, Testing mobile systems

Text Books:

1.	Software Testing, Principles and Practices, Naresh Chauhan, Oxford.
2.	Software Testing- Yogesh Singh, CAMBRIDGE.

Reference Books:

1.	Foundations of Software testing, Aditya P Mathur, 2ed, Pearson.
2.	Software testing techniques – BarisBeizer, Dreamtech, second edition.
3.	Software Testing, Principles, techniques and Tools, M G Limaye, TMH
4.	Effective Methods for Software testing, Willian E Perry, 3ed, Wiley

e-Resources:

1.	https://www.tutorialspoint.com/software_testing_dictionary/test_tools.html

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3110	**PC**	**--**	**--**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

COMPUTER NETWORKS LAB

(For CSE)

Course Objectives:

1	Understand and apply different network commands
2.	Analyze different networking functions and features for implementing optimal solutions
3.	Apply different networking concepts for implementing network solution
4.	Implement different network protocols
5.	Understand and apply different network commands

Course Outcomes:

At the end of the course Students will be able to

1.	Implement datalink layer framing methods like error control and flow control.	K3
2.	Develop client-server applications using sockets.	K4
3.	Examines and implement the various Routing algorithms	K4

LIST OF EXPERIMENTS

1	Implement the data link layer framing methods such as character stuffing and bit stuffing.
2	Implement on a data set of characters the three CRC polynomials – CRC-12, CRC-16 and CRC-CCIP.
3	Write a C program to develop a DNS client server to resolve the given hostname
4	Write a TCP client-server communication programby socket functions.
5	Write a client-server application for chat using UDP
6	Write a C program to perform sliding window protocol.
7	Implement Dijkstra's algorithm to compute the shortest path in a graph.
8	Take an example subnet of hosts. Obtain broadcast tree for it.
9	Take an example subnet graph with weights indicating delay between nodes. Now obtain Routing table art each node using distance vector routing algorithm
10	Get the MAC or Physical address of the system using Address Resolution Protocol?

Reference Books:

1	Computer Networks, Andrew S. Tanenbaum, David J. Wetherall, Pearson Education India; 5th edition, 2013.
2	Data Communication and Networking , Behrouz A. Forouzan, McGraw Hill, 5th Edition,2012

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3111	CS	--	--	3	1.5	20	30	3 Hrs.

WEB TECHNOLOGIES LAB

(For CSE)

Course Objectives:

1	Learn the core concepts of both the frontend and backend programming course
2.	Get familiar with the latest web development technologies
3.	Learn all about PHP and SQL databases
4.	Learn complete web development process
5.	Learn the core concepts of both the frontend and backend programming course

Course Outcomes:

At the end of the course Students will be able to

		KL
1.	Design innovative and interactive web pages using HTML, CSS, and JAVASCRIPT	K4
2.	Develop web pages using XML and AJAX	K4
3.	Design dynamic Web Applications using PHP & MySQL	K5

LIST OF EXPERIMENTS

1 Design Time Table using HTML Table tag.

2 Design the following static web pages required for an online book store website:

HOMEPAGE:

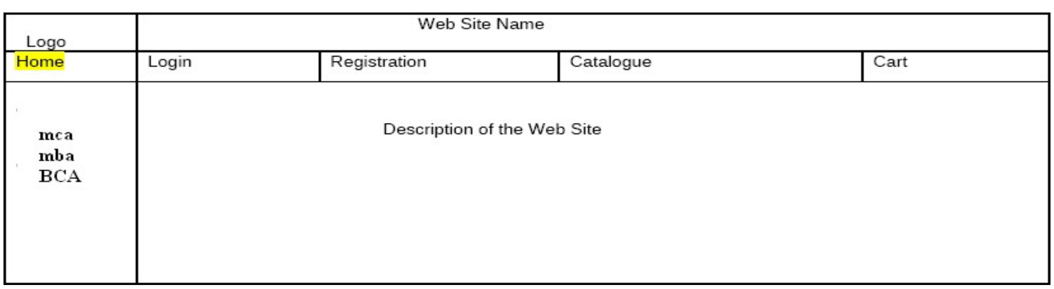
The static home page must contain three **frames**.

Top frame: Logo and the college name and links to Homepage, Login page, Registration page, Catalogue page and Cart page (the description of these pages will be given below).

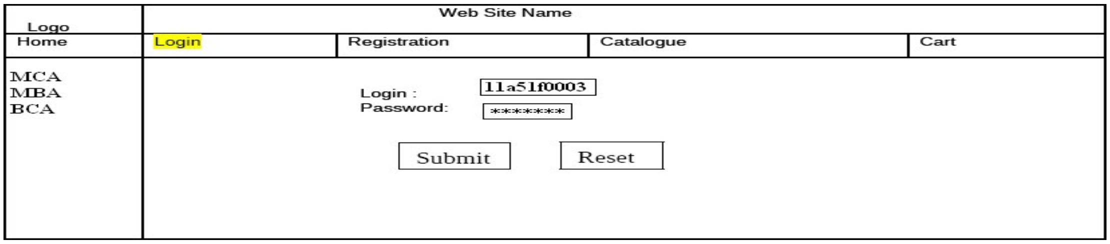
Left frame: At least four links for navigation, which will display the catalogue of respective links.

For e.g.: When you click the link **"MCA"** the catalogue for MCA Books should be displayed in the Right frame.

Right frame: The *pages to the links in the left frame must be loaded here*. Initially this page contains description of the website.



a. LOGINPAGE:

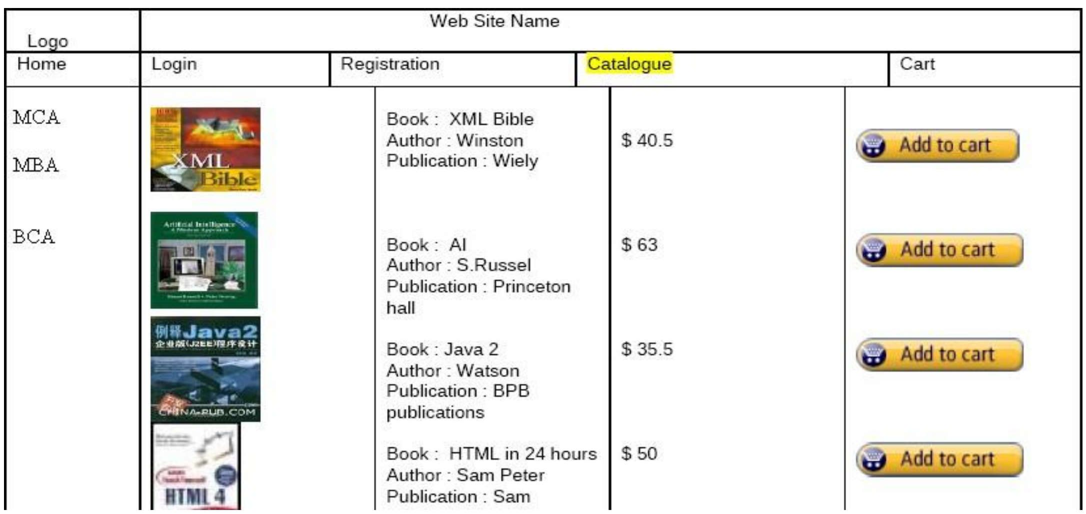


b. CATOLOGUEPAGE:

The catalogue page should contain the details of all the books available in the website in a table: The details should contain the following:

- Snapshot of Cover Page.
- Author Name.
- Publisher.
- Price.

c. Add to cart button.



(d).REGISTRATIONPAGE:

Create a "*registration form* "with the following fields

1) Name (Text field) 2) Password (password field)
3) E-mailid (text field) 4) Phone number (text field)
5) Sex(radio button) 6)Date of birth(3selectboxes)

Languages known (checkboxes–English, Telugu, Hindi, Tamil)

Address (text area)

3	Design a web page using **CSS (C**ascading **S**tyle **S**heets**)** which includes the following: • Inline style sheets • Internal style sheets • External style sheets
4	Design a web page using various CSS Selectors.

5	Design a dynamic web page with validation using JavaScript.
6	Design a HTML having a text box and four buttons Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate JavaScript function should be called to display • Factorial of that number • Is it Armstrong Number or not • Fibonacci series up to that number • Prime numbers up to that number • Is it palindrome or no
7	Write JavaScript programs on Event Handling • Change color of background when the user clicks on the button. • Display calendar for the month and year selected from combo box • On Mouse over event
8	Write an XML file which will display the Book information which includes the following: 1)Title of the book 2)Author Name 3)ISBN number 4) Publisher name 5) Edition 6)Price a. Write a Document Type Definition (DTD) to validate the above XML file. b. Write a XML Schema Definition (XSD) to validate the above XML file
9	Create Web pages using AJAX.
10	Install MySQL Database and create your own database and create tables in it.
11	Write a PHP Program to insert data into the database.
12	Write a PHP Program to validate the user login page.
13	Write a PHP Program to delete data from the database.
14	Write a PHP Program to update data in the database.
15	Write a PHP Program to retrieve data from the database.
Reference Books:	
1	Programming the World Wide Web, 7th Edition Robet W Sebesta, Pearson, 2013.
2	WebTechnologies,1st Edition 7th impression, Uttam K Roy, Oxford, 2012.
3	Pro Mean Stack Development,1st Edition, ELadElrom, A press O'Reilly,2016
4	Java Script & jQuery the missing manual, 2nd Edition, David sawyer mcfarland, O'Reilly, 2011.
5	Web Hosting for Dummies, 1st Edition, Peter Pollock, John Wiley & Sons, 2013.
6	RESTful web services, 1st Edition, Leonard Richardson, Ruby, O'Reilly, 2007.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3112	**PC**	**--**	**--**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

DATA MINING LAB

(For CSE)

Course Objectives:

1	To understand the mathematical basics quickly and covers each and every condition of data mining in order to prepare for real-world problems
2.	The various classes of algorithms will be covered to give a foundation to further applyknowledge to dive deeper into the different flavors of algorithms
3.	Students should aware of packages and libraries of R and also familiar with functions used in R for visualization
4.	To enable students to use R to conduct analytics on large real life datasets
5.	To familiarize students with how various statistics like mean median etc and data can be collected for data exploration in R

Course Outcomes:

At the end of the course Students will be able **to**

1.	Extend the functionality of R by using add-on packages	K3
2.	Examine data from files and other sources and perform various data manipulation tasks on them	K4
3.	Use R Graphics and Tables to apply and visualize results of various statistical operations on data	K3
4.	Apply the knowledge of R gained to data analytics and data mining of real-life datasets	K3

LIST OF EXPERIMENTS

1	Implement all basic R commands.
2	Interact data through .csv files (Import from and export to .csv files).
3	Get and Clean data using swirl exercises. (Use ‘swirl’ package, library and install that topic from swirl).
4	Visualize all Statistical measures (Mean, Mode, Median, Range, Inter Quartile Range etc., using Histograms, Boxplots and Scatter Plots).
5	Create a data frame with the following structure. EMP ID EMP NAME SALARY STARTDATE 1 Satish 5000 01-11-2013 2 Vani 7500 05-06-2011 3 Ramesh 10000 21-09-1999 4 Praveen 9500 13-09-2005 5 Pallavi 4500 23-10-2000 a. Extract two column names using column name. b. Extract the first two rows and then all columns. c. Extract 3rdand 5th row with 2nd and 4th column.
6	Write R Program using ‘apply’ group of functions to create and apply normalization function on

	each of the numeric variables/columns of iris dataset to transform theminto i. 0 to 1 range with min-max normalization. ii. a value around 0 with z-score normalization.
7	Create a data frame with 10 observations and 3 variables and add new rows and columns to it using ‘rbind’ and ‘cbind’ function.
8	Write R program to implement linear and multiple regression on ‘mtcars’ dataset to estimate the value of ‘mpg’ variable, with best R2and plot the original values in ‘green’ and predicted values in ‘red’.
9	Implement k-means clustering using R.
10	Implement k-medoids clustering using R.
11	implement density based clustering on iris dataset.
12	implement decision trees using ‘readingSkills’ dataset.
13	Implement decision trees using ‘iris’ dataset using package party and ‘rpart’.
14	Use a Corpus() function to create a data corpus then Build a term Matrix and Reveal word frequencies.

Reference Books:

1) R and Data Mining: Examples and Case Studies, 1st ed, Yanchang Zhao, Sprnger, 2012.
2) R for Everyone, Advanced Analytics and Graphics, 2nd ed, Jared Lander, Pearson, 2018.

e-Resources:

1) www.r-tutor.com

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--

EMPLOYABILITY SKILLS - I

Common to All Branches

Part-A: Verbal and Soft Skills-I

Course Objectives:

1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2.	To familiarize the learner with high frequency words as they would be used in their professional career.
3.	To inculcate logical thinking in order to frame and use data as per the requirement
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5.	To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:
At the end of the course, Students will be able to

S.No	Outcome	Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.	K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.	K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.	K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.	K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.	K3

SYLLABUS

UNIT-I	**Grammar: (VA)** Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

UNIT-II	**Vocabulary: (VA)** Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.
UNIT-III	**Reasoning: (VA)** Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms.
UNIT-IV	**Usage: (VA)** Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	**Soft Skills:** Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners„s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal„s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from ‗IMS, Career Launcher and Time„ institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.

Part-B: Quantitative Aptitude-I

Course Objectives:

1.	To familiarize students with basic problems on numbers and ratios problems.
2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4.	To inculcate logical thinking by exposing the students to reasoning related questions.
5.	To inculcate logical thinking by exposing the students to reasoning related questions.

Course Outcomes:
At the end of the course, Students will be able to

S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3

SYLLABUS

UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership,

	Sleeping partner concept and problems, Problems on shares and dividends, and stocks.
UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3103	**MC**	**--**	**--**	**3**	**0**	**--**	**--**	**--**

ADVANCED CODING
(Common to CSE & IT)

Course Objectives:

1.	To understand the basics of Java Packages
2	To learn about Collection Framework
3	To investigate different methods of GUI Programming.
4	To learn about Java collections and Libraries

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1.	Solve problems using java collection framework and I/o classes.	K3
2.	Develop multithreaded applications with synchronization.	K3
3.	Develop applets for web applications	K3
4.	Design GUI based applications	K4

SYLLABUS

UNIT-I (10 Hrs)	Standard Library templates and Java collections Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.
UNIT-II (10 Hrs)	The Collections Framework (java.util)- Collections overview, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque. Accessing a Collection via an Iterator, Using an Iterator, The For-Each alternative, Map Interfaces and Classes, Comparators, Collection algorithms, Arrays, The Legacy Classes and Interfaces- Dictionary, Hashtable ,Properties, Stack, Vector.
UNIT-III (10 Hrs)	More Utility classes, String Tokenizer, Bit Set, Date, Calendar, Random, Formatter, Scanner.
UNIT-IV (8 Hrs)	GUI Programming with Swing – Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout. Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events.
UNIT-V (12 Hrs)	Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, Applets – Applets and HTML, Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JCombo Box, Swing Menus, Dialogs.

References:	
1.	Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd.
2.	Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education
3	An Introduction to programming and OO design using Java, J. Nino and F.A. Hosch, John Wiley & sons
4	Introduction to Java programming, Y. Daniel Liang, Pearson Education.
5	Object Oriented Programming through Java, P. Radha Krishna, University Press.
6	Programming in Java S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
7	Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.
8	https://www.geeksforgeeks.org/
9	https://www.tutorialspoint.com/

COMPUTER SCIENCE & ENGINEERING

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

II-SEMESTER

(With effect from **2019-2020**Admitted Batch onwards**)**

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 HS 3201	Managerial Economics and Financial Accountancy	HS	3	3	--	--	25	75	100
B19 CS 3201	Compiler Design	PC	3	3	--	--	25	75	100
B19 CS 3202	UML & Design Pattern	PC	3	3	--	--	25	75	100
B19 CS 3203	Artificial Intelligence	PC	3	3	--	--	25	75	100
B19 CS 3204	MOOCs-I	PE	3	3	--	--	25	75	100
#OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 CS 3205	UML Lab	PC	1.5	--	--	3	20	30	50
B19 CS 3206	Data Analysis and Visualization using Python Lab	PC	1.5	--	--	3	20	30	50
B19 CS 3207	Industrial Training/ Internship/Research Projects in National Laboratories/ Academic Institutions	PR	1	--	--	--	--	50	50
B19 MC 3201	Employability Skills-II	MC	0	3	--	--	--	--	--
B19 MC 3204	Competitive Coding	MC	0	--	--	3	--	--	--
		TOTAL	**22**	**21**	**--**	**9**	**190**	**560**	**750**

#OE-I	Student has to study one Open Elective offered by CE or ECE or EEE or ME or EM&H from the list enclosed.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B19HS3201	HS	3	--	--	3	25	75	3 Hrs.

MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY

(For CSE)

Course Objectives: Students are expected to learn

1.	To Study Managerial Economics and Demand Analysis
2.	To familiarize about the Concepts of Cost and Break-Even Analysis.
3.	To understand the nature of markets and to know the Pricing Policies
4.	To learn about accounting cycle and preparation of Financial Statements.
5.	To know the concept of Capital and sources of raising and Depreciation

Course Outcomes: At the end of the course the student will be able to

S. No	Outcome	KL
1.	Equip oneself with the knowledge of estimating the Demand and demand elasticities for a product.	K2
2.	Have knowledge of Cost and its types and ability to calculate BEP	K3
3.	Understand the nature of different markets	K2
4.	Understand Pricing Practices prevailing in today's business world	K2
5.	Prepare Financial Statements and know how to calculate Profit & Loss for a firm	K3
6.	Know Types of capital and their sources and know how to calculate Depreciation	K2

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Managerial Economics and demand Analysis:** **Managerial Economics:** Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics. **Demand Analysis:** Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of demand forecasting and its Methods.
UNIT-II (10 Hrs)	**Cost Analysis**: Importance of cost analysis, **Types of Cost-** Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; **Elements of costs** – Material, Labour, Expenses; **Methods of costing** - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. **Break-even analysis:** Determination of Breakeven point - Applications, Assumptions and Limitations of Break -even analysis (Theory only).
UNIT-III (10 Hrs)	**Introduction to Markets & Pricing Policies** **Market Structures:** Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. **Pricing:** Importance of pricing and its meaning;**Methods of Pricing**: **Cost Based** -Full cost, Mark-up, Marginal &Break-even; **Demand Based** - Penetrating, Skimming; **Competition Based-** Going rate, Sealed Bid, Discount; **Internet Pricing** - Flat-rate, Usage sensitive.

UNIT-IV (8 Hrs)	**Introduction to Financial Accounting:** Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (outlines only).
UNIT-V (12 Hrs)	**Capital & Depreciation:** Types of Capital - Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, medium term and Long term. **Depreciation -** Meaning, Importance and causes of depreciation; Methods of Depreciation- Straight line and diminishing balancing methods (Theory only)

Text Books:

1.	A R Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N.Appa Rao, Dr.P. Vijayakumar:Managerial Economics and Financial Analysis', Cengage Publications, New Delhi

Reference Books:

1.	Dr.B.Kuberudu& T.V. Ramana : Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K. Gupta & R.K. Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Code	Category	L	T	P	C	I.M	E.M	Exam
B19 CS 3201	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

COMPILER DESIGN

(For CSE)

Course Objectives:

1.	To study the various phases in the design of a compiler
2.	To understand the design of top-down, bottom-up parsers and syntax directed translation schemes.
3.	To introduce LEX and YACC tools.
4.	To learn to develop algorithms to generate code for a target machine.

Course Outcomes:
At the end of the course Students will be able to

1.	Apply the knowledge in different phases of Compiler, specifying different types of tokens in lexical analyzer and use the Compiler tools like LEX.	K3
2.	Construct top down parser, shift reduce parser for CFG.	K3
3.	Construct LR parsers for CFG and summarize Syntax directed translation schemes	K3
4.	Summarize intermediate code and illustrate code optimization techniques	K2
5.	Illustrate code generation and runtime environments	K2

SYLLABUS

UNIT-I (10 Hrs)	**Language Processors:** Introduction to Language Processing, Structure of a Compiler, The Science of Building a Compiler, Compiler-Construction Tools. **Lexical Analysis**: The Role of Lexical Analysis, Input Buffering, Specification of Tokens, Recognitions of Tokens, The Lexical Analyzer Generator LEX.
UNIT-II (10 Hrs)	**Syntax Analysis**: The Role of a Parser, CFG (Definition of CFG, Derivations and Parse Trees, Ambiguity), Writing Grammar (Eliminating Ambiguity, Elimination of Left Recursion and Left Factoring in CFG), Bottom-up Parsing (Shift Reduce Parsing). Top-down Parsing (Recursive-Descent Parsing and Predictive Parsing),
UNIT-III (10 Hrs)	**Introduction to LR Parser** (Simple LR Parsing), More Powerful LR Parser (Canonical LR and LALR Parsing), Using Ambiguous Grammars, Error Recovery in LR parser, The Parser Generator YACC. **Syntax-Directed Translation**: Syntax Directed Definitions (Inherited and Synthesized Attributes, Evaluating an SDD at Nodes of Parse Tree), Evolution Order of SDTS (Dependency Graphs, Ordering the Evaluation of Attributes, S-Attributed Definitions and L-Attributed Definitions), Application of SDTS (Construction of Syntax Trees), Syntax Directed Translation Schemes (Postfix Translation Schemes, Parser Stack Implementation of Postfix SDT's).
UNIT-IV (10 Hrs)	**Intermediate Code Generation**: Variants of Syntax Trees (DAG for Expressions, The Value-Number Method for Constructing DAG's), Three-Address Code (Address and Instructions, Quadruples, Triples), Type Checking (Rules for Type Checking and Type Conversion). **Code Optimization**: The Principal Sources of Optimization, Introduction to Basic Blocks and Flow Graphs, Optimization of Basic Blocks, Introduction to Data-Flow Analysis.

UNIT-V (10 Hrs)	**Code Generation**: Issues in the Design of a Code Generator, The Target Language, A Simple Code Generator, Code Generation from DAG, Peephole Optimization, Register Allocation and Assignment. **Runtime Environments**: Storage Organization, Stack Allocation of Space, Heap Management, Symbol Tables (Symbol Table Per Scope, Use of Symbol Tables).

Text Books:

1.	Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman, Pearson, Pearson Education India; 2nd edition, 2013
2.	Compiler Construction-Principles and Practice, Kenneth C Louden, Cengage Learning, 1 January 200

Reference Books:

1.	Modern compiler implementation in C, Andrew W Appel, Revised edition, Cambridge University Press.
2.	The Theory and Practice of Compiler writing, J. P. Tremblay and P. G. Sorenson, TMH
3.	Writing compilers and interpreters, R. Mak, 3rd edition, Wiley student edition.

e-Resources:

1.	https://nptel.ac.in/courses/106/104/106104123

Code	Category	L	T	P	C	I.M	E.M	Exam
B19 CS 3202	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

UML & DESIGN PATTERNS

(For CSE)

Course Objectives:

1.	To understand the fundamentals of object modeling
2.	To understand and differentiate Unified Process from other approaches
3.	To design with static UML diagrams
4.	To design with the UML dynamic and implementation diagrams
5.	To improve the software design with design patterns

Course Outcomes:
At the end of the course Students will be able to

1.	Illustrate software development process	K2
2.	Design system UML Diagrams	K3
3.	Identify different system behavioral modeling techniques	K2
4.	Select suitable creational pattern while designing a system	K3
5.	Select suitable Behavioural pattern while designing a system	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to UML:** Importance of modeling, principles of modeling, object oriented modeling, conceptual model of the UML, Architecture, Software Development Life Cycle. Structural Modeling: Classes, Relationships, common Mechanisms, and diagrams. Advanced classes, advanced relationships, Object diagrams: common modeling techniques.
UNIT-II (10 Hrs)	**Behavioral Modeling:** Interactions, Interaction diagrams. Use cases, Use case Diagrams, Activity Diagrams, Events and signals, state machines, state chart diagrams.
UNIT-III (10 Hrs)	**Advanced Behavioral Modeling:** Architectural Modeling: Components, Deployment, Component diagrams and Deployment diagrams, Common modeling techniques for component and deployment diagrams. Design Pattern: Introduction, Design Patterns in Smalltalk MVC, Describing Design Patterns, The Catalog of Design Patterns, Organizing the Catalog, How Design Patterns Solve Design Problems, How to Select a Design Pattern, Using a Design Pattern.
UNIT-IV (10 Hrs)	**Creational Patterns:** Abstract Factory, Builder, Factory Method, Prototype, Singleton Structural Patterns: Adapter, Bridge, Composite, Decorator, Façade, Flyweight, Proxy.
UNIT-V (10 Hrs)	**Behavioral Patterns:** Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, Strategy, Template Method, What to Expect from Design Patterns

Text Books:

1.	The unified Modeling language user guide by Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education,Eighth Impression-2009

2.	Design Patterns (Elements of reusable object oriented software), Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, Pearson Education , First Impression-2012
Reference Books:	
1.	Object Oriented Analysis and Design, Satzinger, CENGAGE
2.	Meilir Page-Jones: Fundamentals of Object-Oriented Design in UML, Pearson Education.
e-Resources:	
1.	https://www.tutorialspoint.com/design_pattern/design_pattern_quick_guide.html

Code	Category	L	T	P	C	I.M	E.M	Exam
B19 CS 3203	**PC**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

ARTIFICIAL INTELLIGENCE

(For CSE)

Course Objectives:

1.	To learn about AI problem, Production Systems and their characteristics.
2.	To understand the importance of search and the corresponding search strategies for solving AI problem.
3.	To introduce to Planning, Natural Language Processing and Expert Systems

Course Outcomes:
At the end of the course Students will be able to

1.	Summarize different AI problems and their characteristics.	K2
2.	Describe state space representation for solving AI problems.	K2
3.	Apply optimal, heuristic search strategies for solving AI problems.	K3
4.	Interpret the given facts to different knowledge representational schemes.	K3
5.	Apply AI problem solving approaches to uncertainty, NLP, Planning and Expert Systems.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Artificial Intelligence:** Artificial Intelligence, AI Problems, AI Techniques, Defining the Problem as a State Space Search, Water jug problem, 8-puzzle problem, Tic-Tac Toe problem, Travelling Salesmen Problem, Turing Test, BFS, DFS, Problem Characteristics, Production Systems.
UNIT-II (10 Hrs)	**Search Techniques:** Issues in The Design of Search Programs, Heuristic Search Techniques: Generate-And- Test, Hill Climbing, Best-First Search, A* Algorithm, Problem Reduction, AO*Algorithm, Constraint Satisfaction, Means-Ends Analysis.
UNIT-III (10 Hrs)	**Symbolic Logic:** Propositional Logic, First Order Predicate Logic: Representing Instance and is-a Relationships, Computable Functions and Predicates, Unification & Resolution, Natural Deduction. **Reasoning under Uncertainty:** Introduction to Non-Monotonic Reasoning, Truth Maintenance Systems, Logics for Non-Monotonic Reasoning
UNIT-IV (10 Hrs)	**Knowledge Representation using Rules:** Procedural Vs Declarative Knowledge, Logic programming, Forward Vs Backward Reasoning, Matching Techniques, Partial Matching, RETE Matching, PROLOG. Bayesian Networks and Fuzzy Logic, Dempster-shafer theory, Fuzzy Sets, Crisp Sets, Fuzzy inference and Fuzzy System. **Structured Representations of Knowledge**: Semantic Nets, Partitioned Semantic Nets, Frames, Conceptual Dependency and Scripts.

UNIT-V (10 Hrs)	**Planning** Components of a Planning System, Goal Stack Planning, Non-linear Planning using Constraint Posting, Hierarchical Planning. **Natural Language Processing**: Steps in the Natural Language Processing, Syntactic Processing and Augmented Transition Nets, Semantic Analysis. **Experts Systems:** Overview of an Expert System, Architecture of Expert Systems, Different Types of Expert Systems.

Text Books:	
1.	Artificial Intelligence, Elaine Rich and Kevin Knight, Tata McGraw-Hill Publications, 3rd Edition, Year-2010
2.	Introduction To Artificial Intelligence & Expert Systems, Patterson, PHI publications, First Edition, Year-2015
Reference Books:	
1.	Artificial Intelligence, George F Luger, Pearson Education Publications, 5th Edition, Year-2008
2.	Artificial Intelligence: A modern Approach, Russell and Norvig, Printice Hall, 3rd Edition, Year-2015
3.	Artificial Intelligence, Robert Schalkoff, McGraw-Hill Publications, 3rd Edition, Year-2002
4.	Artificial Intelligence and Machine Learning, Vinod Chandra S.S., AnandHareendran S, First Edition, Year-2014

Code	Category	L	T	P	C	I.M	E.M	Exam
B19 CS 3204	**PE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

MOOCs-I

Professional Electives-II

(For CSE)

MOOCs-I course under Professional Elective –II should belong to the concerned B.Tech. Programme and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration.

The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	**OE**	**3**	**--**	**--**	**3**	**25**	**75**	**3 Hrs.**

OPEN ELECTIVE-I

(For CSE)

Student has to study one Open Elective offered by CE or ECE or EEE or ME or EM&H from the list enclosed.

Offered by	Course Code	Course Name	Offered to
CIVIL ENGINEERING	B19CEOE01	Disaster Management	**CSE,ECE,EEE, IT &ME**
	B19CEOE02	Remote Sensing &GIS	
ELECTRONICS & COMMUNICATIONENGINEERING	B19ECOE01	Basic Electronics	**CE,CSE,EEE, IT &ME**
	B19ECOE02	Signals & Systems	
ELECTRICAL & ELECTRONICS ENGINEERING	B19EEOE01	Introduction To Electrical Systems	**CE,CSE,ECE, IT &ME**
	B19EEOE02	Electrical Estimation and Costing	
MECHNAICALENGINEERING	B19MEOE01	Operations Research	**CE,CSE,ECE, EEE & IT**
	B19MEOE02	Operations Management	
	B19MEOE03	Total Quality Management	
ENGINEERING MATHEMATICS & HUMANITIES	B19BSOE01	Computational statistics with R	**ALL BRANCHES**
	B19BSOE02	Fuzzy sets and Fuzzy Logic	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3205	**PC**	**0**	**0**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

UML Lab

(For CSE)

Course Objectives:

1.	To know the practical issues of the different object-oriented analysis and design concepts
2.	Inculcate the art of object-oriented software analysis and design
3.	Apply forward and reverse engineering of a software system
4.	Carry out the analysis and design of a system in an object-oriented way

Course Outcomes:
At the end of the course Students will be able to

1.	Know the syntax of different UML diagrams.	K2
2.	Create use case documents that capture requirements for a software system	K3
3.	Create class diagrams that model both the domain model and design model of a software system	K3
4.	Create interaction diagrams that model the dynamic aspects of a software system	K3
5.	Write code that builds a software system	K2
6.	Develop simple applications	K4

SYLLABUS

Experiment 1:	Familiarization with Rational Rose or Umbrella environment
Experiment 2:	a) Identify and analyze events b) Identify Use cases c) Develop event table
Experiment 3:	a) Identify & analyze domain classes b) Represent use cases and a domain class diagram using Rational Rose c) Develop CRUD matrix to represent relationships between use cases and problem domain classes
Experiment 4:	a) Develop Use case diagrams b) Develop elaborate Use case descriptions & scenarios c) Develop prototypes (without functionality)
Experiment 5:	a) Develop system sequence diagrams and high-level sequence diagrams for each use case b) Identify MVC classes / objects for each use case c) Develop Detailed Sequence Diagrams / Communication diagrams for each use case showing d) interactions among all the three-layer objects
Experiment 6:	a) Develop detailed design class model (use GRASP patterns for responsibility assignment) b) Develop three-layer package diagrams for each case study

Experiment 7:	a) Develop Use case Packages b) Develop component diagrams c) Identify relationships between use cases and represent them d) Refine domain class model by showing all the associations among classes
Experiment 8:	Develop sample diagrams for other UML diagrams - state chart diagrams, activity diagrams and deployment diagram.

Reference Books:

1.	The unified Modeling language user guide by Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education,Eighth Impression-2009
2.	Design Patterns (Elements of reusable object oriented software), Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, Pearson Education , First Impression-2012

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3206	**PC**	**0**	**0**	**3**	**1.5**	**20**	**30**	**3 Hrs.**

DATA ANALYSIS AND VISUALIZATION USING PYTHON LAB

(For CSE)

Course Objectives:

1.	To acquire programming skills in Python package NumPy and perform mathematical and statistical operations.
2.	To understand the fundamentals of the Pandas library in Python
3.	To develop skills in handling data and also develop basic skills in data analysis andvisualization

Course Outcomes:
At the end of the course Students will be able to

1.	Understand the workings of various numerical techniques, different descriptive measures of Statistics, to solve the engineering problems	K2
2.	Understand how to apply some linear algebra operations to n-dimensional arrays, best practices for creating basic charts	K2
3.	Use NumPy perform common data wrangling and computational tasks in Python	K3
4.	Use Pandas to create and manipulate data structures like Series and Data Frames.	K3
5.	Work with arrays, queries, and data frames, Query Data Frame structures for cleaning and processing and manipulating files	K3

SYLLABUS

Experiment 1:	NumPy Installation using different scientific python distributions(Anaconda, Python(x,y), WinPython, Pyzo)
Experiment 2:	NumPy Basics (np.array, np.arrange, np.linespace, np.zeros, np.ones, np.random.random, np.empty)
Experiment 3:	Arrays (array.shape, len(array), array.ndim, array.dtype, array.astype(type), type(array))
Experiment 4:	Array Manipulation (np.append, np.insert, np.resize, np.delete, np.concatenate, np.vstack, np.hstack)
Experiment 5:	Mathematical Operations(np.add, np.substract, np.divide, np.multiply, np.sqrt, np.sin, np.cos, np.log, np.dot, np.roots) , Statistical Operations(np.mean, np.median, np.std, array.corrcoef( ))
Experiment 6:	NumPy data types
Experiment 7:	NumPyndarray
Experiment 8:	NumPy String Operations

Experiment 9:	NumPy Financial functions
Experiment 10:	NumPy Functional Programming
Experiment 11:	Write a Python program for the following · Importing matplotlib, · Simple Line Plots, · Adjusting the Plot: Line Colors and Styles, Axes Limits, · Labeling Plots, · Simple Scatter Plots, · Histograms, · Customizing Plot Legends, · Choosing Elements for the Legend, · Multiple Legends, · Customizing Colorbars, · Multiple Subplots, · Text and Annotation, · Customizing Tick.
Experiment 12:	1) Python Programs for Interacting with data in text format, Interacting with data in MongoDB, Cleaning, Filtering and Merging Data A) Pandas DataSeries: 1) Write a Pandas program to create and display a one-dimensional array-like object containing an array of data using Pandas module. 2) Write a Pandas program to convert a Panda module Series to Python list and it's type. 3) Write a Pandas program to add, subtract, multiple and divide two Pandas Series. 4) Write a Pandas program to convert a NumPy array to a Pandas series. Sample Series: NumPy array: [10 20 30 40 50] Converted Pandas series: 0 10 1 20 2 30 3 40 4 50 dtype: int64 B) Pandas DataFrames: Consider Sample Python dictionary data and list labels: exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily', 'Michael',

	'Matthew', 'Laura', 'Kevin', 'Jonas'], 'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19], 'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1], 'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']} labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'] 1) Write a Pandas program to create and display a DataFrame from a specifieddictionary data which has the index labels. 2) Write a Pandas program to change the name 'James' to 'Suresh' in name column of the DataFrame. 3) Write a Pandas program to insert a new column in existing DataFrame. 4) Write a Pandas program to get list from DataFrame column headers. 5) Write a Pandas program to get list from DataFrame column headers. C) Pandas Index: 1) Write a Pandas program to display the default index and set a column as an Index in agivendataframe. 2) Write a Pandas program to create an index labels by using 64-bit integers, using floating-point numbers in a given dataframe. D) Pandas String and Regular Expressions: 1) Write a Pandas program to convert all the string values to upper, lower cases in agivenpandas series. Also find the length of the string values. 2) Write a Pandas program to remove whitespaces, left sided whitespaces and right sidedwhitespaces of the string values of a given pandas series. 3) Write a Pandas program to count of occurrence of a specified substring in a DataFrame column. 4) Write a Pandas program to swap the cases of a specified character column in a givenDataFrame. E) Pandas joining and merging DataFrame: 1) Write a Pandas program to join the two given dataframesalong rows and assign alldata. 2) Write a Pandas program to append a list of dictionaries or series to a existing DataFrame and display the combined data. 3) Write a Pandas program to join the two dataframes with matching records from both sides where available. F) Pandas Time Series: 1) Write a Pandas program to create a) Datetime object for Jan 15 2012. b) Specific date and time of 9:20 pm. c) Local date and time. d) A date without time. e) Current date.

f) Time from a datetime.
g) Current local time.

G) Pandas Grouping Aggregate:

Consider dataset:

school	class	name	date_Of_Birth	age	height	weight	address
s001	V	Alberto Franco	15/05/2002	12	173	35	street1
s002	V	Gino Mcneill	17/05/2002	12	192	32	street2
s003	VI	Ryan Parkes	16/02/1999	13	186	33	street3
s001	VI	Eesha Hinton	25/09/1998	13	167	30	street1
s002	V	Gino Mcneill	11/5/2002	14	151	31	street2
s004	VI	David Parkes	15/09/1997	12	159	32	street4

1) Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.
2) Write a Pandas program to split the following dataframe by school code and getmean, min, and max value of age for each school.

H) Pandas Styling:

1) Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black.
2) Create a dataframe of ten rows, four columns with random values. Write a Pandasprogram to highlight the maximum value in each column.
3) Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight dataframe's specific columns.

I) Excel:

1) Write a Pandas program to import excel data into a Pandas dataframe.
2) Write a Pandas program to find the sum, mean, max, min value of a column of file.

J) Plotting:

1) Write a Pandas program to create a horizontal stacked bar plot of opening,

	closing stock prices of any stock dataset between two specific dates. 2) Write a Pandas program to create a histograms plot of opening, closing, high, low stock prices of stock dataset between two specific dates. 3) Write a Pandas program to create a stacked histograms plot of opening, closing, high,low stock prices of stock dataset between two specific dates with more bins. K) Pandas SQL Query: 1) Write a Pandas program to display all the records of a student file. 2) Write a Pandas program to select distinct department id from employees file.

Reference Books:	
1.	Wesley J Chen , Core Python Programming 2nd Edition, Kindle Edition
2.	Jake VanderPlas, Python Data Science Handbook: Essential Tools for Working with Data 1st Edition, Kindle Edition, Oreilly Publisher
3.	Mark Summerfield, Programming in Python 3--A Complete Introduction to the Python Language, Second Edition, Additson Wesley
4.	Phuong Vo.T.H , Martin Czygan, Getting Started with Python Data Analysis, Packt Publishing Ltd
5.	Ivan Idris, Python Data Analysis, Packt Publishing Ltd
6.	Magnus VilhelmPersson and Luiz Felipe Martins, Mastering Python Data Analysis, Packt Publishing Ltd

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CS3207	**PR**	**0**	**0**	**0**	**1**	**--**	**50**	**3 Hrs**
INDUSTRIAL TRAINING/INTERNSHIP/ RESEARCH PROJECTS IN NATIONAL LABORATORIES/ ACADEMIC INSTITUTIONS								
CSE								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	**MC**	**3**	**--**	**--**	**--**	**--**	**--**	**--**

EMPLOYABILITY SKILLS II

Common to All Branches

Part-A: Verbal and Soft Skills-II

Course Objectives:

1.	To expose the students to bettering sentence expressions and also forming equivalents.
2.	To instill reading and analyzing techniques for better comprehension of written discourses
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4.	To inculcate conversational skills, nuances required when interacting in different situations
5.	To build/refine the professional qualities/skills necessary for a productive career and to in still confidence through attitude building.

Course Outcomes: The students will be able to

S.No	Outcome	Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.	K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences	K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).	K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.	K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.	K3

SYLLABUS

UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author„s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.
UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1.	Guide to IELTS, Cambridge University Press.
2.	Barron„s GRE guide
3.	Newspapers like The Hindu, Times of India, Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN

Text Books:

1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.

Reference Books:

1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie

5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.

Part-B: Quantitative Aptitude-II

Course Objectives: The objective of introducing quantitative aptitude –II is:

1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3.	To familiarize the students with finding out accurate date and time related problems
4.	To enable the students solve the puzzles using logical thinking.
5.	To expose the students to various problems based on geometry and mensuration.

Course Outcomes:
At the end of the course students will be able to

S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3

SYLLABUS

UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.
UNIT-III	Periods, Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles
UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi
3.	Puzzles to puzzle you by shakunataladevi
4.	More puzzles by shakunataladevi
5.	Puzzles by George summers

Reference Books:	
1.	Barron„s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from ‗IMS, Career Launcher and Time„ institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3204	**MC**	**--**	**0**	**3**	**0**	**--**	**--**	**--**

COMPETITIVE CODING

(Common to CSE & IT)

Course Objectives:

1.	To understand the essentials of competitive coding.
2.	To learn the use of Standard Template Library components.
3.	To learn about selection-based algorithms.
4.	To learn different pattern matching and Graph algorithms.

Course Outcomes:
At the end of the course Students will be able to

1.	Use Mathematical functions to solve coding tasks.	K3
2.	Apply STL functions to solve recursive algorithms.	K3
3.	Solve coding tasks related to selection based problems.	K3
4.	Apply Pattern matching and Graph algorithms to solve various problems.	K3
5.	Use Mathematical functions to solve coding tasks.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Competitive Coding** Introduction to Competitive coding and coding Platforms. Coding solution Vs. Efficient Coding solution. Types of solution approaches. Analyzing problem specific data requirement, Various data representations. Essential Data structures for fast coding. Various Syntactical I/O techniques comparison. Numbers, operations (including exponentiation). Integer properties (positive, negative, even, odd, divisible, prime, etc), Fractions, percentages and ratios. Point, vector, Cartesian coordinates (2D integer grid).
UNIT-II (10 Hrs)	**Essentials to Competitive coding** Asymptotic Notations – (Big-O), Evaluating the runtime complexity – Space Complexity - Towers-of-Brahma – Standard Template Libraries - Square root functions, primality testing and related techniques. Euclidean algorithms. Recursion techniques. Organizing data in O(nlogn). Binary search techniques. Basic Techniques: Dynamic Arrays, Set structures, Map structures, Iterators and ranges, Generating Subsets, Generating permutations, Backtracking techniques, Pruning the search. Bit masking.
UNIT-III (10 Hrs)	**Essential Coding Algorithms** Selection based algorithms: sorting, Coin change problem, Fractional selections, Schedules matching, Activity marking, heap sort, Huffman coding techniques, Minimizing sums, Data compression. Finding method count, Subsequence and related problems, paths in grid. DP with Bit mask.

UNIT-IV (10 Hrs)	**String & Tree Algorithms** Naive string searching, z-algorithm, Manacher's algorithm, Rabin-Karp, KMP Algorithm, Suffix arrays. Tree Traversals, Diameter, All longest paths, Applying search property to tree structures. Fenwick tree and Segment Tree.
UNIT-V (10 Hrs)	**Graph Algorithms** Graph Algorithms: Graph Traversals, Shortest paths: Dijkstra's algorithm ,Bellman-Ford Algorithm ,Floyd Warshall Algorithm - Adjacency List Representation – Euler path, tour , cycle – Eulerian Graph - Johnson"s Algorithm for All-pairs shortest path – Shortest path in Directed Acyclic Graph. Bridges and articulation points. Strongly connected components in directed graphs. 2-SAT.

Reference Books:

1.	Data Structures and Algorithm Analysis in C++ – Mark Allen Weiss, Third Edition,Pearson Edu Publishers, 2014.
2.	Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, first edition, Tata Mc Graw Hill Publishers.2008.
3.	Advanced Data Structures – Peter Brass, first edition, Cambridge University Press, 2008.
4.	Introduction to Algorithms by ThomosH.cormen, 3rd Edition, PHI,2009.
5.	https://www.geeksforgeeks.org
6.	https://www.tutorialspoint.com